Derivation of Motion Tracking Equation

The fundamental equation for motion tracking is derived from the 'Brightness Constancy Assumption'. This assumption states that the brightness (intensity) of a particular point on an object remains constant as it moves from one frame to the next, even if it's position changes.

Let $I(x, y, t)$ be the intensity of pixel at location (x, y) at time t.

Let the point move by a small displacement (dx, dy) over a small time interval dt.

So, $I(x, y, t) = I(x + dx, y + dy, t + dt)$ — (i)

Derivation:

We expand the right side of eqn (i) using a first-order Taylor series

$$I(x+dx, y+dy, t+dt) \approx I(x,y,t) + \frac{\partial I}{\partial x} dx + \frac{\partial I}{\partial y} dy + \frac{\partial I}{\partial t} dt$$

putting this into eqn (i) we get,

$$I(x,y,t) = I(x,y,t) + \frac{\partial I}{\partial x} dx + \frac{\partial I}{\partial y} dy + \frac{\partial I}{\partial t} dt$$

$$\Rightarrow \frac{\partial I}{\partial x} dx + \frac{\partial I}{\partial y} dy + \frac{\partial I}{\partial t} dt = 0$$

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$$\Rightarrow \frac{\partial I}{\partial x}\frac{dx}{dt} + \frac{\partial I}{\partial y}\frac{dx}{dt} + \frac{\partial I}{\partial t} = 0$$

[divided by dt]

$$\Rightarrow \frac{\partial I}{\partial x}u + \frac{\partial I}{\partial y}v + \frac{\partial I}{\partial t} = 0 \quad \left[\text{let } u = \frac{dx}{dt},\ v = \frac{dy}{dt}\right]$$

Now tracking problem between two frames:

Given two consecutive frames,

frame 1 = $I(x, y, t)$

frame 2 = $I(x, y, t+1)$.

the goal is to estimate the motion vector (u, v) for each pixel.

However, this equation provides only one equation with two unknowns, so the motion cannot be equally determined. This is known as the aperture problem.

To solve this, motion is estimated within a local neighborhood window, assuming nearby pixels have similar motion.

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To solve this we use Lucas-Kanada method, which assumes the motion (u, v) is constant within a small window around the target pixel.

Step 1: compute Gradients:

$$\frac{\partial I}{\partial x} = I_x \approx \frac{I(x+1, y, t) - I(x-1, y, t)}{2}$$

$$\frac{\partial I}{\partial y} = I_y \approx \frac{I(x, y+1, t) - I(x, y-1, t)}{2}$$

$$\frac{\partial I}{\partial t} = I_t \approx I(x, y, t+1) - I(x, y, t)$$

for a window of n pixels, we have n equations:

$$I_{x_i} u + I_{y_i} v + I_{t_i} = 0$$

$$\Rightarrow I_{x_i} u + I_{y_i} v = -I_{t_i}$$

In matrix form $Ad = b$:

$$\begin{bmatrix} I_{x_1} & I_{y_1} \\ I_{x_2} & I_{y_2} \\ \vdots & \vdots \\ I_{x_n} & I_{y_n} \end{bmatrix} = \begin{bmatrix} -I_{t_1} \\ -I_{t_2} \\ \vdots \\ -I_{t_n} \end{bmatrix}$$

Step 2: solve via least squares:

$$\begin{bmatrix} u \\ v \end{bmatrix} = (A^T A)^{-1} A^T b$$

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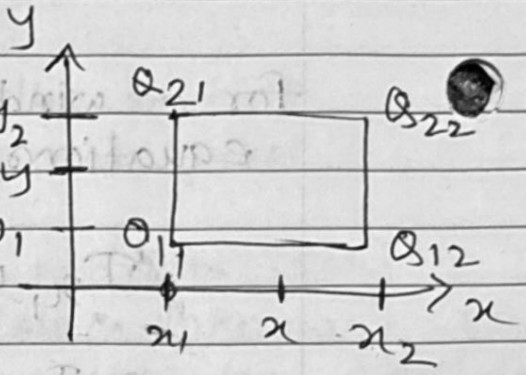
This expands to the standard Lucas-Kanada solution:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \Sigma I_x^2 & \Sigma I_x I_y \\ \Sigma I_x I_y & \Sigma I_y^2 \end{bmatrix}^{-1} \begin{bmatrix} -\Sigma I_x I_t \\ -\Sigma I_y I_t \end{bmatrix}$$

Bilinear Interpolation Derivation

Bilinear interpolation estimates the value of a function $f(x, y)$ at a non-integer coordinate (x, y) using the four nearest integer neighbors:

$Q_{11} = (x_1, y_1)$
$Q_{12} = (x_1, y_2)$
$Q_{21} = (x_2, y_1)$
$Q_{22} = (x_2, y_2)$



let x be between x_1, x_2
and y y_1, y_2

Normalize co-ordinates such that,

$x_1 = 0$ $\quad$ $y_1 = 0$
$x_2 = 1$ $\quad$ $y_2 = 1$

The local offsets are

$$\alpha = x - x_1$$
$$\beta = y - y_1$$

① Step 1: Interpolation at X direction:

find value R_1 at (x, y_1) between Q_{11} and Q_{21}:

$$f(x, y_1) \approx (1-\alpha) f(x_1, y_1) + \alpha f(x_2, y_1)$$

find value R_2 at (x, y_2) between Q_{12} and Q_{22}:

$$f(x, y_2) \approx (1-\alpha) f(x_1, y_2) + \alpha f(x_2, y_2)$$

② Step 2: Interpolate at Y:

find value P at (x, y) between R_1 and R_2

$$f(x, y) \approx (1-\beta) f(x, y_1) + \beta f(x, y_2)$$

③ Step 3: Combine:

$$f(x, y) \approx (1-\beta)[(1-\alpha) f(Q_{11}) + \alpha f(Q_{21})] + (1-\alpha)\beta f(Q_{12}) + \alpha f(Q_{22}).$$

So final formula:

$$f(x, y) = (1-\alpha)(1-\beta) I(0,0) + \alpha(1-\beta) I(1,0) + (1-\alpha)\beta I(0,1) + \alpha\beta I(1,1)$$

This is a weighted avg where the weights are the areas of the rectangles opposite the corner points